Last Mile Connectivity

Version 0 — Static Last Mile Assignment Model

1 Problem Description

We consider a static last-mile connectivity problem in which vehicles operate on predefined soft routes rather than strict door-to-door service. This is a single time-period static optimization problem: all passenger requests and vehicle positions are known beforehand.

Scenario Description

- All vehicles transport passengers to their destination (e.g., a metro station).
- Each vehicle operates along a soft-route corridor—a flexible path from its origin toward the destination.
- Vehicles are allowed to take small detours within a predefined radius to pick up nearby passengers.
- Each vehicle has a fixed seating capacity.

2 Sets

- Passengers: $P = \{1, 2, \ldots, p\}$
- Vehicles: $V = \{1, 2, \ldots, v\}$
- Stops: $S = \{1, 2, \ldots, s\}$

- Soft routes: $R = \{1, 2, \ldots, r\}$

3 Parameters

Passenger Parameters

- o_p: origin location
- $S_p \subseteq S$: feasible pickup stops
- $l_p^{\max}$: maximum walking distance
- l_p^{ideal}: ideal walking distance
- d_p^{walk}: walking distance of passenger p
- T_p^{ideal}: ideal in-vehicle trip duration

Vehicle Parameters

- Cap_v: vehicle capacity
- R_v: base soft route
- $\Delta_v^{\max}$: maximum allowable detour
- C_v^{km}: operating cost per km

Weight Parameters

- γ: vehicle operating distance weight
- α: walking discomfort weight
- β: passenger trip duration weight
- μ: occupancy encouragement weight
- λ: rejection penalty weight

Stop Parameters

- loc_s: stop location

Derived Parameters

- δ_v: total passenger detour for vehicle v
- B_v: base route distance (origin $\rightarrow$ destination without detours) for vehicle v

4 Decision Variables

Passenger–Vehicle Assignment

$$x_{pv} = \begin{cases} 1 & \text{if passenger } p \text{ is assigned to vehicle } v, \\ 0 & \text{otherwise.} \end{cases}$$

Rejection Penalty Indicator

$$r_p = \begin{cases} 1 & \text{if passenger } p \text{ is NOT assigned,} \\ 0 & \text{otherwise.} \end{cases}$$

Passenger Stop Assignment

$$x_{ps}^{\text{pickup}} = \begin{cases} 1 & \text{if } p \text{ stops at } s, \\ 0 & \text{otherwise.} \end{cases} \qquad x_{ps}^{\text{drop}} = \begin{cases} 1 & \text{if } p \text{ is dropped at } s, \\ 0 & \text{otherwise.} \end{cases}$$

Vehicle Route Deviation δ_v total additional distance travelled by vehicle v compared to its base route, due to pickup and drop-off operations.

5 Constraints

Hard Constraints If service is mandatory:

$$\sum_{v \in V} x_{pv} = 1 \quad \forall p \in P$$

If rejection is allowed (but penalised):

$$\sum_{v \in V} x_{pv} \leq 1 \quad \forall p \in P, \qquad r_p = 1 - \sum_{v \in V} x_{pv} \quad \forall p \in P$$

Vehicle capacity:

$$\sum_{p \in P} x_{pv} \leq \mathrm{Cap}_v \quad \forall v \in V$$

Passenger–stop linking:

$$\sum_{s \in S} x_{ps}^{\mathrm{pickup}} = \sum_{v \in V} x_{pv}, \qquad \sum_{s \in S} x_{ps}^{\mathrm{drop}} = \sum_{v \in V} x_{pv} \quad \forall p \in P$$

Route deviation bound:

$$\delta_v \leq \Delta_v^{\max}$$

Soft Constraints

$$\begin{aligned} \mathrm{WalkingPenalty}_p &= \alpha \max\bigl(0,\ d_p^{\mathrm{walk}} - l_p^{\mathrm{ideal}}\bigr) \\ \mathrm{TripPenalty}_p &= \beta \max\bigl(0,\ T_p - T_p^{\mathrm{ideal}}\bigr) \\ \mathrm{OccupancyReward}_v &= -\mu(\mathrm{Cap}_v - n_v) \end{aligned}$$

6 Objective Function

$$\min\ \gamma \sum_{v \in V} C_v^{\mathrm{km}}(B_v + \delta_v) + \alpha \sum_{p \in P} \max(0, d_p^{\mathrm{walk}} - l_p^{\mathrm{ideal}}) + \beta \sum_{p \in P} \max(0, T_p - T_p^{\mathrm{ideal}}) - \mu \sum_{v \in V} (\mathrm{Cap}_v - n_v) + \lambda \sum_{p \in P} r_p$$

7 Greedy Initialization Algorithm

Input Passenger set P; vehicle set V; stop set S; predefined soft routes R_v; capacities Cap_v; maximum deviation $\Delta_v^{\max}$; walking distances d_{ps}; travel time and route information.

Output Assignments x_{pv}, x_{ps}^{pickup}, x_{ps}^{drop} and constructed routes for each vehicle.

Step 1: Initialization

For each vehicle $v \in V$: set occupancy $n_v = 0$, current route = base route R_v, total deviation $\delta_v = 0$, assigned passengers = $\emptyset$. Let $P_{\mathrm{unassigned}} = P$.

Step 2: Candidate Pool Construction

For each vehicle v, construct a feasible candidate set $C_v \subseteq P_{\text{unassigned}}$. A passenger p is included in C_v if: a feasible pickup stop $s^{\text{pick}} \in S_p$ exists, a feasible drop-off stop exists on the route, walking constraints are satisfied, and the assignment does not violate hard constraints. From all such passengers, select the top $K \cdot \text{Cap}_v$ candidates based on a preliminary score (e.g., proximity to route or estimated detour).

Step 3: Generate Candidate Subsets (Randomized, Variable Size)

From C_v, generate a collection of candidate subsets of *varying sizes*:

$$\mathcal{S}_v = \{S \subseteq C_v \;\mid\; 1 \leq |S| \leq \text{Cap}_v\}$$

A vehicle is *not* required to be filled to capacity—partial loads are allowed whenever they yield a lower greedy cost (e.g., when forcing additional passengers would cause excessive detour or walking penalties).

Because $|\mathcal{S}_v|$ grows combinatorially, we do not enumerate it exhaustively. Instead, N subsets are drawn via randomized sampling:

1. For $i = 1, \ldots, N$:
    - Sample a subset size $k_i \sim \mathcal{U}\{1, 2, \ldots, \text{Cap}_v\}$ (uniform, or biased toward larger k if higher occupancy is preferred).
    - Sample k_i passengers without replacement from C_v, uniformly or weighted by the preliminary proximity/detour score from Step 2.
    - Add the sampled subset S_i to $\mathcal{S}_v$.
2. Remove duplicate subsets.

The resulting $\mathcal{S}_v$ contains up to N subsets of heterogeneous sizes, which are passed to Step 4.

Step 4: Greedy Cost Evaluation

For each subset $S \in \mathcal{S}_v$, compute:

$$\text{GreedyCost}(v, S) = \gamma \cdot \delta_v(S) + \alpha \sum_{p \in S} d_p^{\text{walk}} + \beta \sum_{p \in S} \hat{T}_p - \mu\big(\text{Cap}_v - |S|\big)$$

where $\delta_v(S)$ is the route deviation when serving all passengers in S, d_p^{walk} is the walking distance, and $\hat{T}_p$ is the estimated in-vehicle travel time of passenger p. The occupancy term $-\mu(\text{Cap}_v - |S|)$ is included so that, under variable subset sizes, the greedy phase is not biased toward trivially small subsets and remains aligned with the global objective. $\delta_v(S)$ is computed by inserting pickup and drop-off stops of all passengers in S into the vehicle route.

Step 5: Feasibility Check

For each subset S, retain only those satisfying the route deviation constraint $\delta_v(S) \leq \Delta_v^{\max}$.

Step 6: Select Best Subset

$$S^* = \arg\min_{S \in \mathcal{S}_v} \text{GreedyCost}(v, S)$$

Step 7: Assign and Update

For all passengers $p \in S^*$: assign passenger p to vehicle v and assign pickup and drop-off stops. Update:

- $x_{pv} = 1$, $x_{ps}^{\text{pickup}} = 1$, $x_{ps}^{\text{drop}} = 1$
- $n_v = |S^*|$ (which may be less than Cap_v)
- Update vehicle route
- Set $\delta_v = \delta_v(S^*)$

Remove all $p \in S^*$ from $P_{\text{unassigned}}$.

Step 8: Repeat

Repeat Steps 2–7 for each vehicle until all vehicles are processed or no feasible passengers remain.

Remark This greedy initialization evaluates combinations of passengers for each vehicle rather than assigning passengers sequentially. Allowing variable-sized subsets lets the algorithm leave a vehicle partially filled whenever doing so improves the greedy cost, while the occupancy term in GreedyCost prevents systematic under-filling. A simplified greedy cost function is used to efficiently approximate the impact of assignments, enabling better selection of passenger groups while maintaining computational tractability.

8 Cost Calculation

After completing the greedy and reshuffling phases, the total system cost of the final solution is:

$$C_{\text{total}} = \gamma \sum_{v \in V} C_v^{\text{km}}(B_v + \delta_v) + \alpha \sum_{p \in P} \max(0, d_p^{\text{walk}} - l_p^{\text{ideal}}) + \beta \sum_{p \in P} \max(0, T_p - T_p^{\text{ideal}}) - \mu \sum_{v \in V} (\text{Cap}_v - n_v) + \lambda \sum_{p \in P}$$

This value represents the objective minimized by the greedy and reshuffling procedures.